Two explicit rank-three geometries for the Fischer extension $\mathrm{Fi}'_{24} : 2$

Human Compression project

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Abstract

We specify a regular polytope of type $\{6, 35\}$ and a regular rank-three hypertope with triangular diagram $(6, 6, 35)$, both with full type-preserving automorphism group $\mathrm{Fi}'_{24} : 2$. A compact word certificate is verified in the faithful degree-306936 Atlas representation. Generation uses explicit elements of orders 23 and 29 together with the retained complete maximal-subgroup classification. All intersection and flag conditions are verified directly in small parabolic subgroups. The construction addresses the Fi'_{24} possible exception in Theorem 5.3 of Piedade–Tranchida's specified preprint edition; no priority claim is made.

1 Scope and retained identification

Write $H = \mathrm{Fi}'_{24}$ and $G = \mathrm{Aut}(H) = H : 2$, also denoted Fi_{24} by the Atlas. We retain the Atlas identification of the two supplied permutations as standard generators of this faithful representation of G, the simplicity of H, $|\mathrm{Out}(H)| = 2$, and the complete maximal-subgroup classification used below. In particular,

$$|H| = 1255205709190661721292800, \qquad |G| = 2510411418381323442585600.$$

The program does not enumerate the order of this ambient permutation group. The named identification and classification are mathematical inputs, not consequences of a few standard-generator tests.

The two primary files are `F24d2G1-p306936B0.g1` and `.g2`, defining permutations c and d, respectively. The first assigns the GAP name `b11` and the second `b21`. The Atlas places c in outer class $2C$ and d in class $8D$. The certificate verifies every identity in the Atlas full-extension standard-generator checker:

$$|c| = 2, \quad |d| = 8, \quad |cd| = 29, \quad |z| = 54, \quad |cz^{27}| = 3, \quad |cd^2| = 20, \qquad z = cdcd^6.$$

Since cd has odd order, its image in $G/H \cong C_2$ is trivial. Thus d, like c, is outer, and every even-length word in $c^{\pm 1}, d^{\pm 1}$ is inner.

The source is arXiv:2605.04646v2, dated 28 May 2026. Theorem 5.3 on physical pages 19–20 concerns groups $H < G \leq \mathrm{Aut}(H)$ and leaves the two Fischer socles as possible exceptions. Our claim concerns the full extension $\mathrm{Fi}'_{24} : 2$, not the simple socle alone. The discussion lies in Section 5.2; Section 5.3 concerns the lamplighter group. This certificate is separate from the ten previously treated sporadic cases and from the $\mathrm{Fi}_{22} : 2$ certificate.

2 Saved words and exact generation

We use $x^h = h^{-1}xh$ and evaluate words from left to right. Signed letters $1, 2$ mean c, d, with negative signs denoting inverses. The complete word file is `F24_compact.g`. Let w_a be its word of length 50 and let $w_1, \ldots, w_8$ be its eight words of length 40. Set

$$a = w_a(c,d)^{21}, \qquad h_j = w_j(c,d), \qquad x_j = cc^{h_j}, \qquad n_j = |x_j|, \qquad z_j = h_j x_j^{(n_j-1)/2}.$$

The checker verifies that every n_j is odd. Bray's elementary centralizer identity shows that each z_j centralizes c. Indeed, if $x = cc^h$ and $n = 2k+1 = |x|$, then $cxc = x^{-1}$ and

$$c^{hx^k} = x^{-k}(cx)x^k = cx^{2k+1} = c.$$

Every such commutation is also checked directly. We use these words only to specify elements; no assertion that they generate the whole centralizer is needed. Define

$$b = (z_7z_1z_7z_1z_7z_7z_6z_3z_8z_1)^{12}, \qquad r = (c,a,b).$$

The checker verifies that the three entries are distinct nonidentity involutions, that $[c,b] = 1$, and that

$$(|ca|, |cb|, |ab|) = (6, 2, 35).$$

The even lengths of the saved words imply $a, b \in H$; this parity information is consistent with the construction, although the generation argument only needs $c \notin H$.

Lemma 1. *A subgroup of G containing c and elements of orders 23 and 29 is G.*

Proof. The complete retained list of maximal subgroups has the following orders. To avoid ambiguity in extension notation, the table gives the exact CTblLib identifiers; `F3+` denotes H and `frob` denotes $29:28$. The Atlas primary list has the same groups; its two 3-local entries of orders 35814871872 and 27549901440 appear in the opposite order. No splitting beyond the displayed identifiers is inferred.

CTblLib maximal-subgroup identifier	Order
`F3+`	1255205709190661721292800
`2xFi23`	8178940946586009600
`2^2.Fi22.2`	516494013235200
`S3xO8+(3):S3`	178278473318400
`O10-(2).2`	50030759116800
`3^7.O7(3):2`	20056328248320
`3^(1+10):(2xU5(2):2)`	9697565306880
`2^12.M24`	1002795171840
`2^2.U6(2):S3x2`	441447874560
`2^(1+12)_+.3_1.U4(3).2^2_{122}`	321052999680
`3^3.[3^10].(L3(3)x2^2)`	35814871872
`3^2.3^4.3^8.(S5x2S4)`	27549901440
`S4xO8+(2):S3`	25082265600
`2^(3+12).(L3(2)xS6)`	3963617280
`2^(7+8).(S3xA8)`	3963617280
`(S3xS3xG2(3)):2`	305690112
`S5xS9`	43545600
`S6xL2(8):3`	1088640
`7:6xS7`	211680
`7^(1+2)_+:(6xS3).2`	24696
`frob`	812

Only the first order is divisible by $23 \cdot 29$. A proper subgroup containing elements of both prime orders would lie in a maximal subgroup conjugate to H, hence in H itself because $H \lhd G$. This contradicts the presence of $c \notin H$. □

The word file supplies an 80-letter word u and a 114-letter word v in r. The exact permutation tests give $|u(r)^2| = 23$ and $|v(r)| = 29$. Thus Lemma 1 proves $\langle r \rangle = G$. The checker compares every retained maximal identifier and order and verifies the divisibility assertion. These checks do not reprove completeness of the maximal-subgroup classification.

Now put $s = (c, a, a^b)$. Its pair orders are verified as $(6, 6, 35)$. Because $|ab| = 35$,

$$a(a^b) = (ab)^2, \qquad b = \big(a(a^b)\big)^{17} a.$$

Consequently $\langle s \rangle = \langle r \rangle = G$. This argument establishes generation; the incidence conditions below are verified independently for each marking.

3 Intersection property, flags and full automorphisms

For either triple q, write $G_I = \langle q_i : i \in I \rangle$ for $I \subseteq \{0, 1, 2\}$. The complete checker constructs all seven proper standard subgroups and verifies the 49 ordered equalities

$$G_I \cap G_J = G_{I \cap J} \qquad (I, J \neq \{0, 1, 2\}). \tag{1}$$

The remaining 15 pairs involve the full set, so their equalities follow from $G_I \leq G$. Thus each triple has the full intersection property.

Let $P_i = G_{\{0,1,2\} \setminus \{i\}}$. For each i, take the other indices $j < k$. We also check the exact set equality

$$(P_j P_k) \cap P_i = (P_j \cap P_i)(P_k \cap P_i). \tag{2}$$

The right side is automatically contained in the left. For the converse, the program enumerates every pair $(u, v) \in P_j \times P_k$ and checks that $uv \in P_i$ implies membership in the right side. These are product sets, not generated subgroups. The total number of pair tests is 1168 for r and 1824 for s; all six equations pass. All subgroups used in these tests are proper rank-two or smaller parabolics, so no ambient enumeration is hidden in this procedure.

Here is the explicit incidence bridge. Use left cosets, with incidence meaning nonempty intersection. Translate an incident triple so its type-i element is P_i. The other two elements have representatives $x, y \in P_i$. Their incidence gives $x^{-1}y \in P_j P_k$. By (2), write $x^{-1}y = uv$ with $u \in P_j \cap P_i$ and $v \in P_k \cap P_i$. Then $xu = yv^{-1}$ is a common representative of all three cosets. Every pairwise incident triple therefore has a common representative. Incident pairs also have common representatives by definition and extend using the third standard coset. Thus all flags extend to chambers and G is flag-transitive.

Equation (1) identifies every residue stabilizer with the corresponding generated subgroup. A panel has two chambers because its singleton subgroup has order two. Every rank-two residue is connected because its two involutions generate its dihedral chamber group. The whole chamber graph is connected because the three involutions generate G. Hence these geometries are thin and residually connected. The common intersection of the P_i is trivial, so G acts faithfully and regularly on chambers. A type-preserving automorphism fixing a chamber fixes its unique adjacent chamber of each type, and then every chamber by connectedness. Every element lies in a chamber, so such an automorphism is trivial. Since G is already chamber-transitive, it is the full type-preserving automorphism group.

Theorem 1. *The saved triples define a regular polytope of type* $\{6, 35\}$ *and a regular rank-three hypertope with triangular diagram* $(6, 6, 35)$*, both with full type-preserving automorphism group* $\mathrm{Fi}'_{24} : 2$*.*

Proof. The generation argument and the exact checks give the geometries just described. For r, the endpoints commute, so the retained string-C-group correspondence supplies the regular polytope. For s, all three pair orders exceed two, giving the nonlinear triangular diagram. The preceding automorphism argument applies to both. □

4 Reproducibility and limitations

The two hash-bound external Atlas permutation files contain 2152215 and 2273175 bytes, respectively. Before execution, the driver verifies that each is exactly one literal permutation assignment, that its cycles have pairwise distinct labels, and that all labels lie in $\{1, \ldots, 306936\}$. Together they contribute 4425390 external data bytes. Their URLs and hashes are recorded; they are not redistributed in this packet. The exact installed CTblLib tables and name, maximal-subgroup and order indices are bound separately.

The proof computation uses the complete saved words, the complete local checker and fixed-witness program, and the Python driver. Search traces and the random construction program are not premises and are not part of the distributed certificate. Negative controls test input syntax and cycle validity, the generation interface, distinctness, intersection property, the independent flag condition and the halving inverse. No Lean declaration or proof-kernel verification of GAP algorithms is claimed. The full programs, driver and external data requirements remain visible; a short word file alone is not the entire computational cost.

The conclusion removes the identified edition's possible exception for this full extension once the explicit retained inputs and computation are accepted. A bounded contemporary literature check is recorded separately in the packet. It is not an exhaustive priority investigation, and we do not claim the first construction or a new result about the simple socle.

References

[1] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 28 May 2026, Theorem 5.3.

[2] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[3] R. A. Wilson and collaborators, *Atlas of Finite Group Representations*, Fischer group Fi'_{24} page, `https://brauer.maths.qmul.ac.uk/Atlas/spor/F24/`.

[4] J. N. Bray, An improved method for generating the centralizer of an involution, *Arch. Math.* **74** (2000), 241–245.

[5] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1; CTblLib 1.3.11. The exact input files are bound in the accompanying ledger.